ITA-0607 MACHINE LEARNING

SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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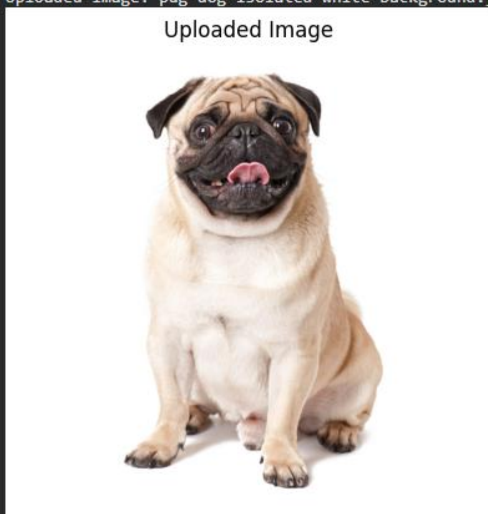
RGB:

```
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.applications.mobilenet_v2 import (
    preprocess_input,
    decode_predictions
)
from tensorflow.keras.preprocessing import image
from google.colab import files
# 1. Load pretrained CNN
model = MobileNetV2(weights="imagenet")
print("CNN model loaded successfully!")
# 2. Upload an animal image
uploaded = files.upload()
filename = list(uploaded.keys())[0]
print("Uploaded image:", filename)
# 3. Display original image
original_image = image.load_img(filename)
plt.imshow(original_image)
plt.title("Uploaded Image")
plt.axis("off")
plt.show()
# 4. Resize image for MobileNetV2
img = image.load_img(
    filename,
    target_size=(224, 224)
)
# 5. Convert image into an array
img_array = image.img_to_array(img)
# 6. Add batch dimension
img_array = np.expand_dims(
    img_array,
    axis=0
)
# 7. Preprocess image
img_array = preprocess_input(img_array)
# 8. Make prediction
```

```
predictions = model.predict(img_array)
# 9. Decode top three predictions
results = decode_predictions(
    predictions,
    top=3
)[0]
# 10. Print predictions
print("\nTop 3 Predictions:")
for number, result in enumerate(results, start=1):
    predicted_name = result[1].replace("_", " ")
    confidence = result[2] * 100
    print(
        f"{number}. {predicted_name}: "
        f"{confidence:.2f}%"
    )
# 11. Get best prediction
best_name = results[0][1].replace("_", " ")
best_confidence = results[0][2] * 100
# 12. Display final result
plt.imshow(original_image)
plt.title(
    f"Prediction: {best_name}\n"
    f"Confidence: {best_confidence:.2f}%"
)
plt.axis("off")
plt.show()
```

OUTPUT:

```
CNN model loaded successfully!
Choose Files pug-dog-iso...kground.jpg
pug-dog-isolated-white-background.jpg(image/jpeg) - 3649342 bytes, last modified: 7/25/2026 - 100% done
Saving pug-dog-isolated-white-background.jpg to pug-dog-isolated-white-background.jpg
Uploaded image: pug-dog-isolated-white-background.jpg
```




```
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 1s/step
```

```
Top 3 Predictions:
1. pug: 70.74%
2. bull mastiff: 4.47%
3. Brabancon griffon: 2.34%
```